

BU5300 – IT Project Management

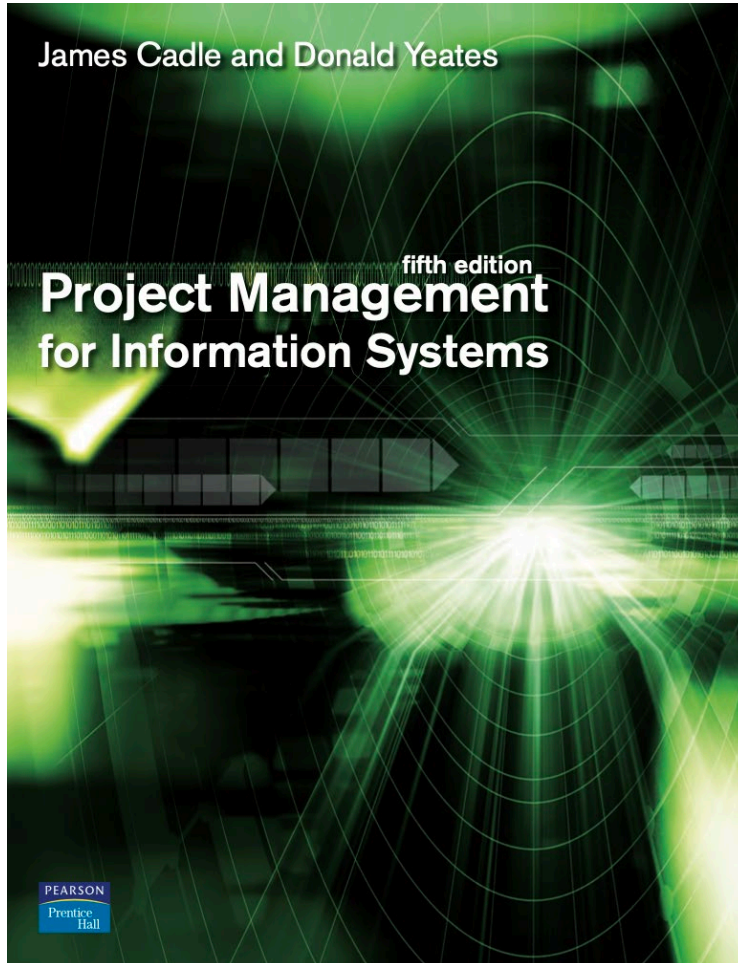
IT Project Management

Lecture 8: **Risk (and Opportunity) Management**

Chapter 15

Overview of the Course

Lectures and Book Chapters



Lecture 1 - Chapters 1 and 2

Lecture 2 - Chapters 3 and 4

Lecture 3 - Chapter 6

Lecture 4 - *Agile Development*

Lecture 5 - Chapters 7 and 8

Lecture 6 - Chapter 9 and 10

Lecture 7 - Chapters 11 - 14

Lecture 8 - Chapter 15

Lecture 9 - Chapters 18 and 21

Lecture 10 - Chapters 20, 22 and 23

IMPORTANT! Changes may occur. Always check on Canvas and TimeEdit

Learning Outcomes

- Describe the risk management process
- List typical project risks
- Assess risks and prepare a risk map
- Create a risk management document
- List typical strategies for risk management

Introduction

Risk Management in IT Projects

-
- Wearable Technology**
Healthcare

What exactly is risk?

- From the latin: *“to dare”*
- A known event that may occur, but we don't know exactly when
- Used in daily speech if something that has only negative consequences if it occurs. (e.g., smart watch services not available during generation of abnormal values of heartbeat).
- But can in principle also mean a positive effect
 - Ex, the **risk** that the IT system has a greater positive effect on the working environment than planned (e.g., security feature may improve product result)

Uncertainty and Risk

- **Uncertainty:** a lack of understanding and awareness of issues, events, paths to follow, or solution to pursue.
- **Risk:** Uncertain **event or condition** that, if **occurs**, has a **positive** or **negative effect** on one or more project objectives



Software project risk definition

- “we define a risk factor as a condition that can present a serious threat to the successful completion of a software development project”
(Schmidt et al., 2001: p. 7)
- Remember: risk factors may also be linked to other types of IT projects

Schmidt, Roy, et al. **Identifying software project risks: An international Delphi study.** Journal of management information systems 17.4 (2001): 5-36.

Identifying Software Project Risks: An International Delphi Study

ROY SCHMIDT, KALLE LYYTINEN,
MARK KEIL, AND PAUL CULE

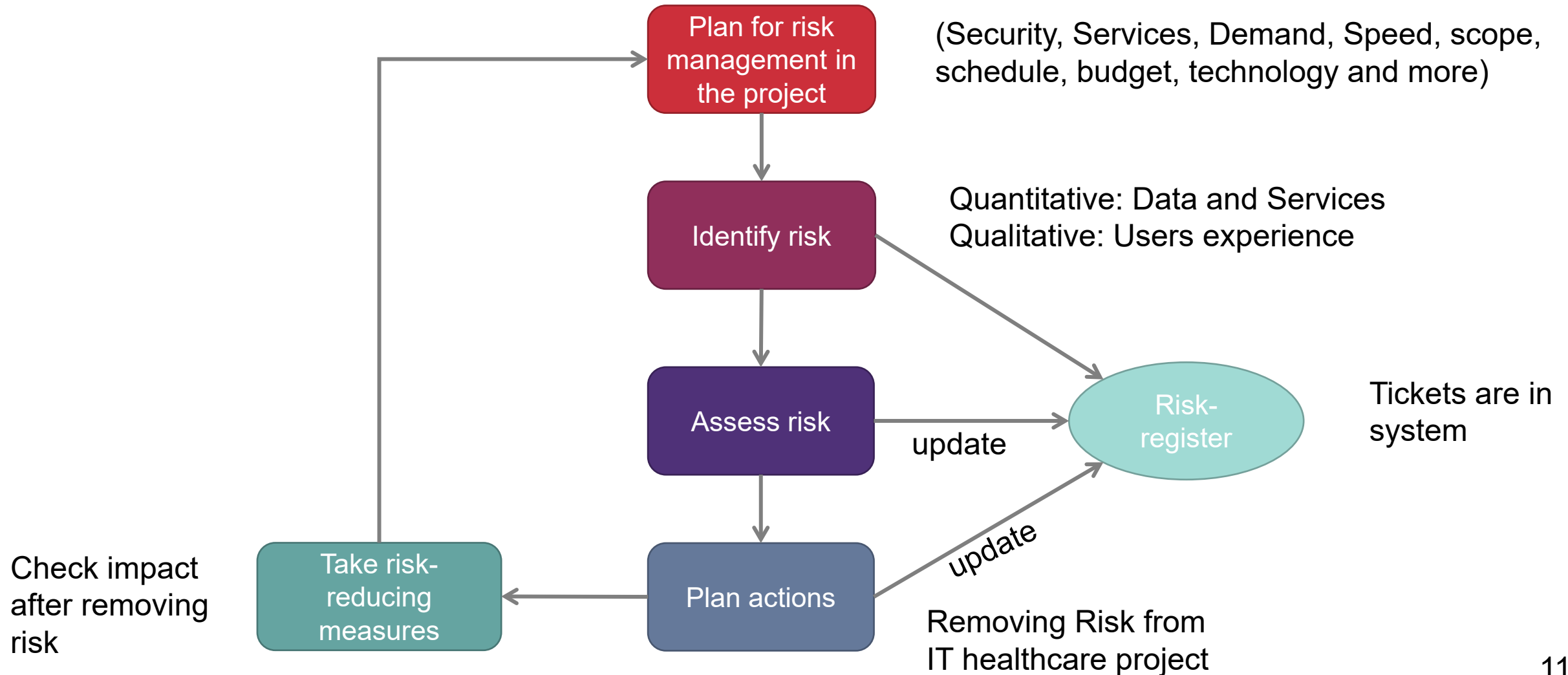
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MARK KEIL is Associate Professor in the Department of Computer Information Systems at Georgia State University. His research focuses on software project management, with particular emphasis on understanding and preventing software project escalation. His research is also aimed at providing better tools for assessing software project risk and removing barriers to software use. Keil's research has been published in *MIS Quarterly*, *Journal of Management Information Systems*, *Sloan Management Review*, *Communications of the ACM*, *IEEE Transactions on Engineering Management*, *Decision Support Systems*, and other journals. He currently serves as coeditor of *Database* and as an Associate Editor for the *MIS Quarterly*.

PAUL CULE is an assistant professor of Management in the College of Business Administration at Marquette University. His research interests are in organizations in transition, information technology, reengineering, software development methods, and management, as well as the roles of IT in changing societies. He has been published in *Communications of the ACM*, *Database*, *Information Systems Management*, and the proceedings of ICIS and AIS. Prior to embarking on an academic career he spent 33 years in industry in Britain, Canada, and the United States. His experience

Risk management in IT projects: the process



Essentials of Risk Management

Risk management requires:

- A mechanisms to keep risks under review and address them
- A means of identifying the potential risks to the project (e.g., Healthcare project)
- An assessment of the likelihood of each risk's materializing (Probability positive or negative)
- An assessment of the probable impact of each risk
- The formulation of measures to avoid each risk's occurring
- The development of fallback measures to upgrade the risks if avoidance actions fail
- The determination of the urgency of the risk and of taking appropriate countermeasures (Prioritize risks, higher, medium and lower)
- Establishing the ownership of risks

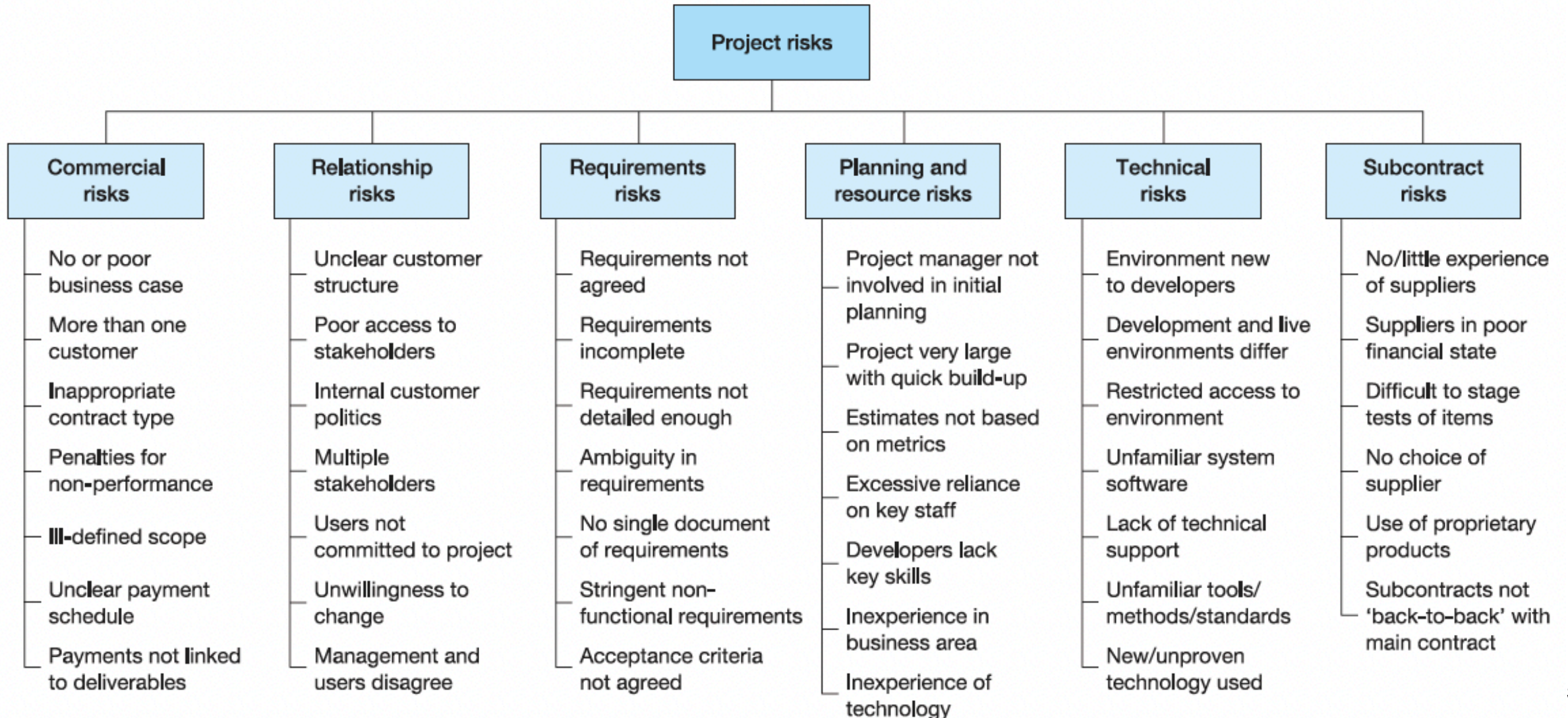
Identify Risks

Risk Management in IT Projects

- Schmidt, Roy, et al. **Identifying software project risks: An international Delphi study.** Journal of management information systems 17.4 (2001): 5-36.



Risk Breakdown Structure



Risk Categories

by Schmidt et al., 2001

Risk categories	Risk factor example
1. Context	Gap between the culture of the business and the changes needed to succeed
2. Ownership	Lack of support from top management
3. Stakeholders	Problems meeting different stakeholders' expectations
4. Change management	Inadequate change management in the project so that the project does not go according to plan
5. Scope	Change in scope and/or objective of the project along the way.
6. Requirements	Too few requirements to freeze on time.
7. Economy	Too little money has been set aside for maintenance.
8. The plan	Unrealistic dates for completion of partial deliveries.
9. Development process	Incorrect development process in relation to organization, expertise, technical solution and type of project.
10. Project participants	Lack of good leadership with a focus on individuals and teams that will be doing the job.
11. Competence	Lack of desirable expertise when the project needs it.
12. Technology	Unstable technological architecture and few other good experiences to build on
13. External dependencies	Insufficient control with consultants, suppliers and subcontractors
14. Planning	Attitude that planning is indispensable

Workshop to identify risk factors in the project

- Brainstorm risk factors
- Read and sort them all into categories
- Discussion
- Document



Assess Risks

Risk Management in IT Projects

Risk Assessment

- The process of risk assessment involves the analysis of their *impact* and *likelihood*
- Typically, both impact and likelihood are assessed at a few qualitative levels (low, medium, high) or on a scale 1 to 5 for example
- We can also assess opportunities in the same way (green for high, red for low)

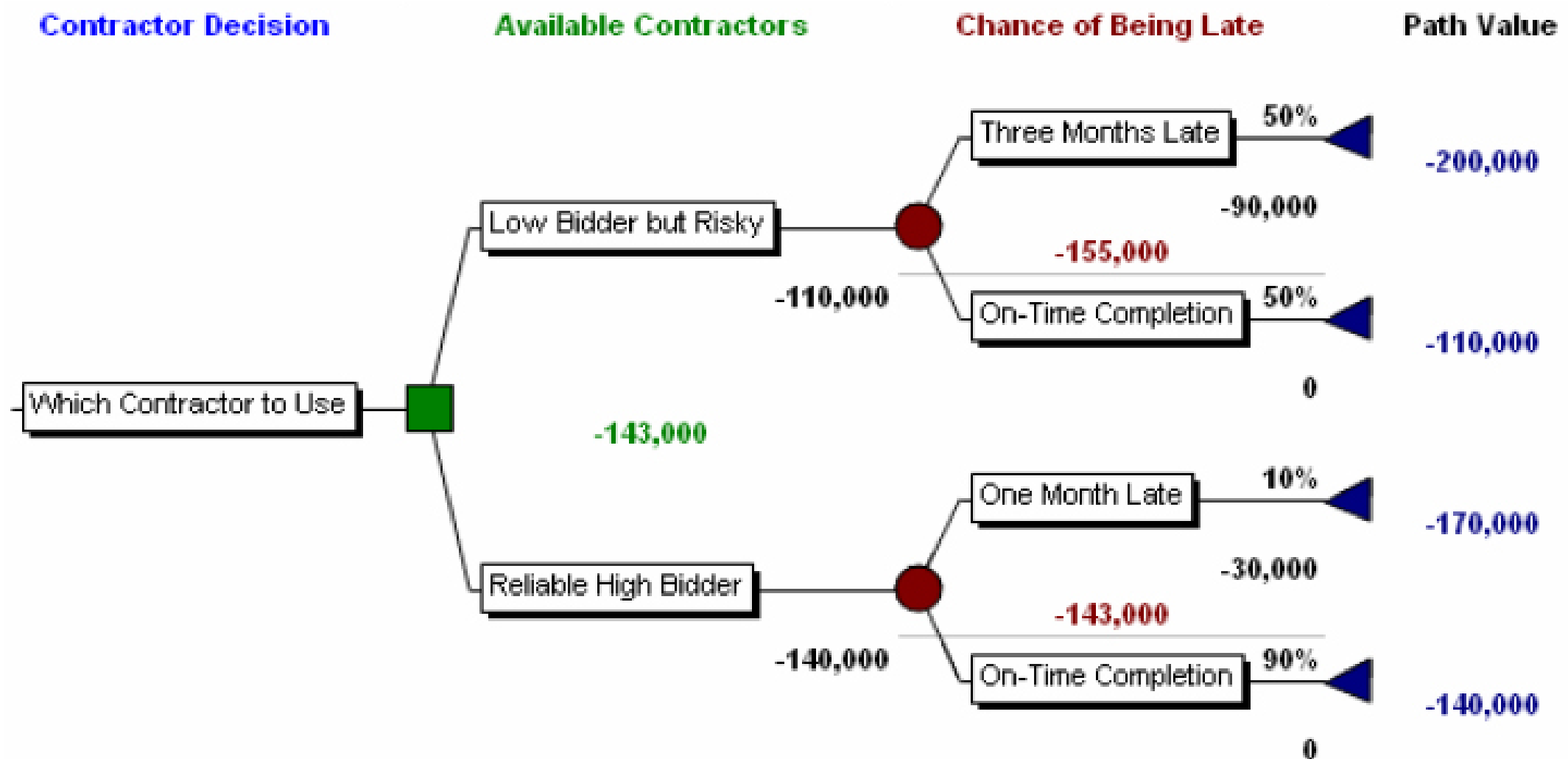
		Impact		
		Low	Medium	High
Probability	High	Low	Medium	High
	Medium	Low	Medium	Medium
	Low	Low	Low	Low

Quantification of risk

- Risk in IT projects is often defined based on the probability of a negative event occurring combined with the severity of the consequence

Risk factor	Probability of occurring (1-5)	Severity rating (1-5)	Evaluation
Project goal and need is not well-defined	3	5	15
Changes in functional requirements	5	4	20
Project deliverables are incompletely defined	1	5	5
Project schedule is not clear	1	3	3
...			

Decision Tree Analysis

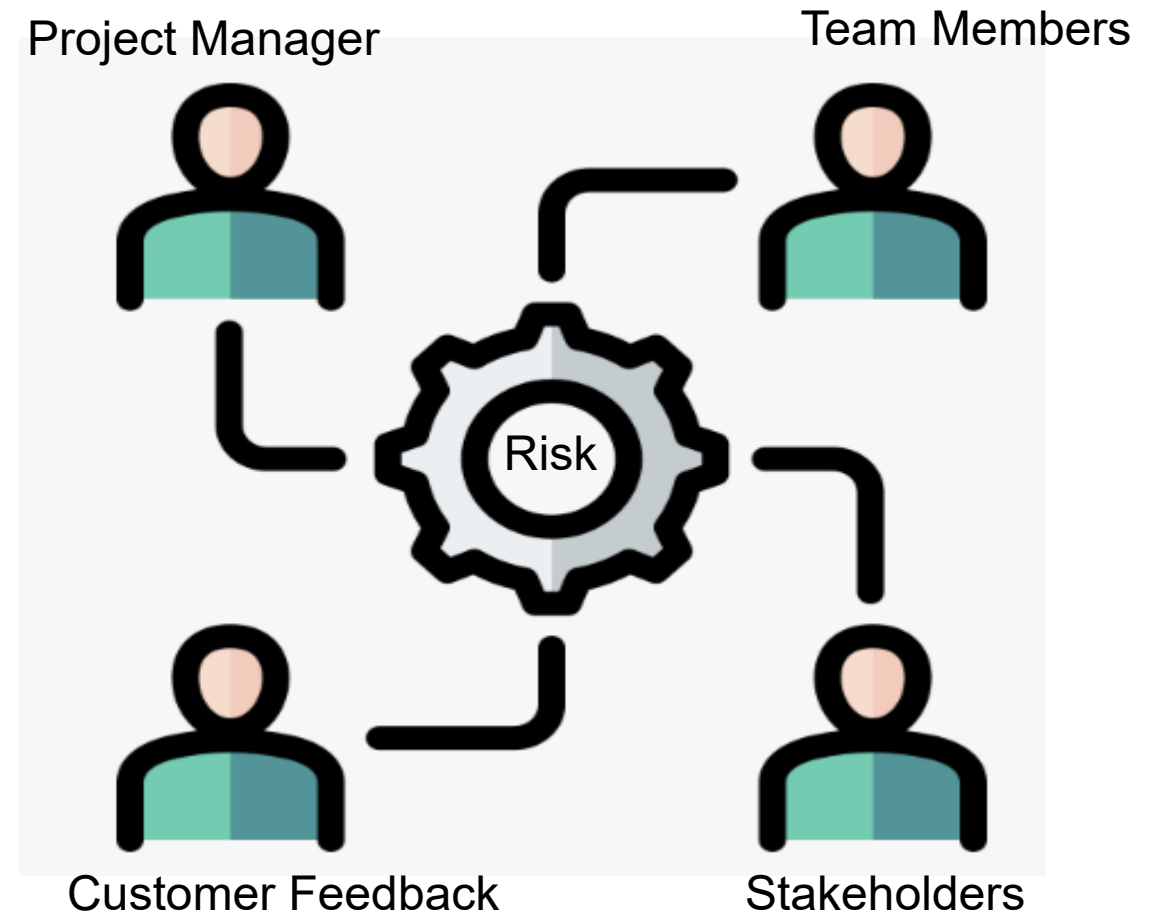


Plan Actions

Risk Management in IT Projects

Plan for risk management in the project

- How will the IT project work on managing risk?
 - Who should update the risk register?
 - Template?
 - Regular meetings?
 - Reporting?
- Important that everyone working on the project is involved in risk management



Strategies for dealing with threats aka Risk Actions or Responses

- **Avoid.** changing the project management plan to eliminate the threat entirely
 - Examples: **extending the schedule**, **changing the strategy**, or **reducing scope**, even **shut down the project entirely**.
 - Some early risks can be avoided by clarifying requirements, obtaining information, improving communication, or acquiring expertise. (Versioning based project).
- **Transfer.** shifting the negative impact of a threat, along with ownership of the response, to a third party
 - For example, when a partner has capabilities that the organization does not possess, it may be prudent to transfer some work and its concurrent risk contractually to the partner.

Strategies for dealing with threats aka Risk Actions or Responses

- **Mitigate.** reduce the probability and/or impact of an adverse risk event to be within acceptable threshold limits
 - For example, designing redundancy into a system may reduce the impact from a failure of the original component. (e.g., reduce the cost of product when demands become less, but still earns stable).
- **Accept.** the project team has decided not to change the project plan to deal with a risk **OR** is unable to identify any other suitable response strategy
 - The most common acceptance strategy is to establish a **contingency reserve**, including time, money, or resources to handle the risks

Risk and knowledge sharing

- Our understanding and assessment of risk depends on those who work closely with the problems/solutions
- Important knowledge sharing across the project and in your organization to reduce risk. (Different team members of different projects can work on different risks to mitigate them).

Further Considerations

Risk Management in IT Projects

Risk Management Plan

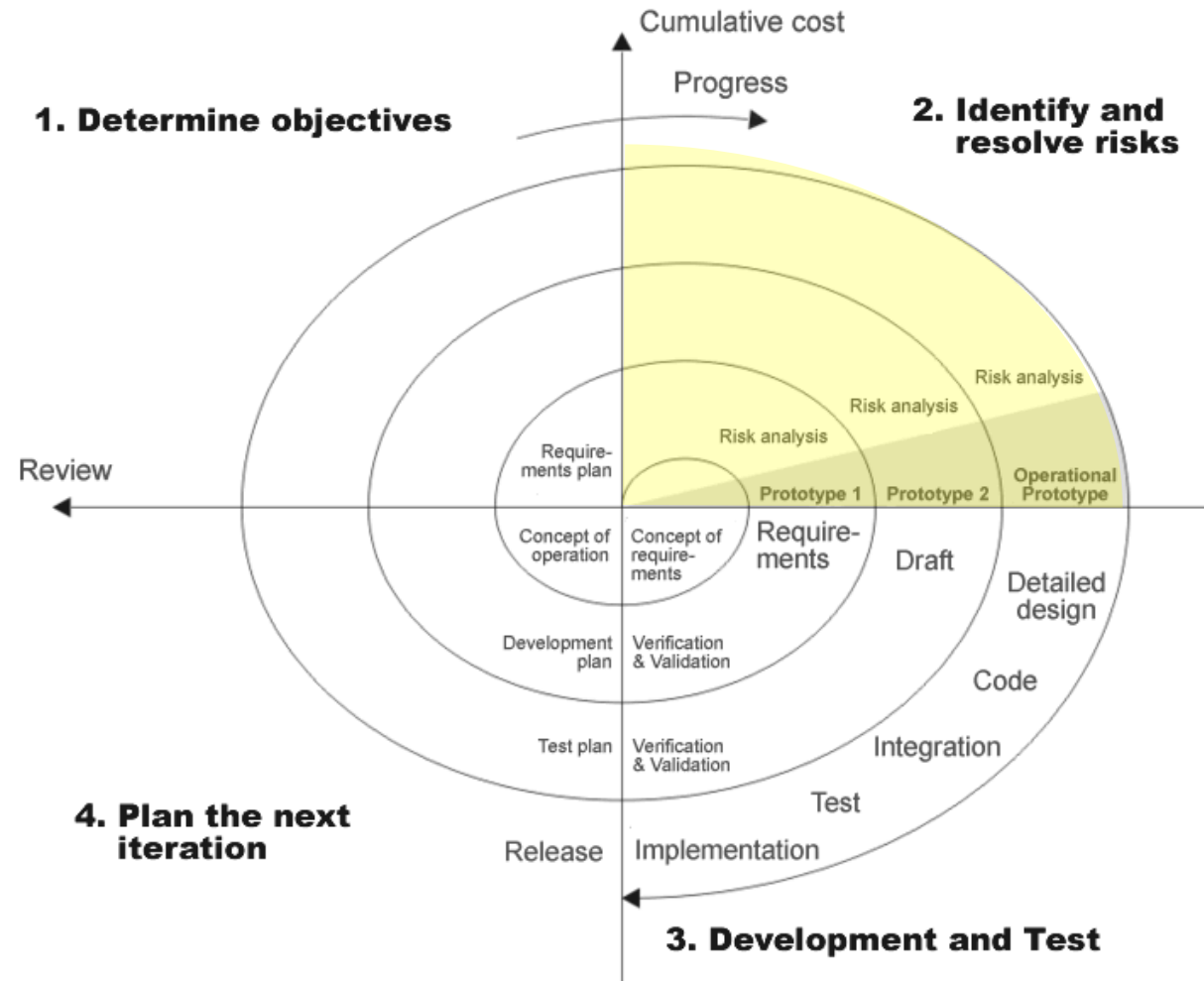
- A list of all risk factors in the project
 - Evaluation
 - Ownership
 - Actions and action log
- Templates
 - PRINCE2: <https://prince2.wiki/management-products/risk-register/>
 - PMBOK: <https://www.stakeholdermap.com/project-templates/risk-management-plan.html>
 - Others?

Risk Ownership

Who is responsible for monitoring and managing a risk factor?

- Part of the process of risk identification is to decide who should be the owner of each risk
- **The Owner should be someone who:**
 - has sufficient information concerning the risk
 - has the necessary resources, and
 - possesses the authority to do something about the risk

The spiral model is based on risk management



Special considerations for Agile

- Adaptive environments tend to exhibit high variability
- High variability → more uncertain risk
- Risk need to be addressed with each iteration or incremental work
 - In selection of work to be completed
 - Risk is identified, analyzed, and managed like regular projects

Lean Risk Management

- Lean Components: **Deliver values, eliminate waste and continous improvement.**
- “Plan” is often a heavyweight document full of hard-to-remember sections
- Risk management plans can be particularly lean with just a few outlined items, such as
 - Responsibilities
 - Risk management approach, and
 - Needed funding resources
- See example of a risk management plan of fault tolerance features compromising performance aspects

1. Objectives	
Determine and reduce the level of risk of the fault tolerance features causing unacceptable performance (e.g., throughput, response time, power consumption). Create a description of and a development plan for a set of low-risk fault tolerance features.	
2. Deliverables and Milestones	
By week 3: Evaluation of fault tolerance options, assessment of reusable components, draft workload characterization, evaluation plan for prototype exercise, and description of prototype. By week 7: Operational prototype with key fault tolerance features, workload simulation, instrumentation and data reduction capabilities. Draft description, plan for fault tolerance features. By week 10: Evaluation and iteration of prototype, revised description and plan for fault tolerance features.	
3. Responsibilities	
• System Engineer: G. Smith assigned to Tasks 1, 3, 4, 9, 11 and to support of Tasks 5, 10 • Lead Programmer: C. Lee assigned to Tasks 5, 6, 7, 10 and to support of Tasks 1, 3 • Programmer: J. Wilson assigned to Tasks 2, 8 and to support of Tasks 5, 6, 7, 10	
4. Approach	
Design-to-Schedule prototyping effort, driven by hypotheses about fault tolerance-performance effects, using multicore processor, real-time OS, add prototype fault tolerance features. Evaluate performance with respect to representative workload. Refine Prototype based on results observed.	
5. Resources	
\$60K	Full-time system engineer, lead programmer, programmer (= 10 weeks)*(3 staff)*(\$2K/staff-week)
\$0K	3 Dedicated workstations (from project pool)
\$0K	2 Target processors (from project pool)
\$0K	1 Test co-processor (from project pool)
\$10K	Contingencies
\$70K	Total

Questions?

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Bibliography

- Cadle, James, and Donald Yeates, eds. **Project management for information systems**. Pearson education, 2004.
- Ruhe, Günther, and Claes Wohlin, eds. **Software project management in a changing world**. Vol. 21. Berlin/Heidelberg, Germany: Springer, 2014.
- Guide, A. **Project management body of knowledge (pmbok® guide)**. Project Management Institute. Vol. 11. 2001.
- Schmidt, Roy, et al. **Identifying software project risks: An international Delphi study**. Journal of management information systems 17.4 (2001): 5-36.
- Hulett, David T. **Decision tree analysis for the risk averse organization**. PMI EMEA Congress. 2006.
- Boehm, Barry W. **Software risk management: principles and practices**. IEEE software 8.1 (1991): 32-41.

Exercise

Create a risk register / document of your designated project

- Identify the most important risks using the checklist by Schmidt et al. (2001) OR top-10 risk item tracking by Boehm (1991)
- Assess the risk factor (likelihood and impact) of each risk item
- Describe at least one mitigation strategy for each item, according to the four strategies for dealing with risks (avoid, transfer, mitigate or accept)



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